# Decoding Blood Relations
# A Modern Approach to Verbal Reasoning

*A Complete Study Guide with Solved Examples from S Chand Academy*

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## Section 1: Conceptual Framework & Methodology

Blood Relations is a foundational topic in competitive examinations including MBA, SSC, Banking, and NDA entrance tests. To solve these problems efficiently, a structured diagrammatic approach is highly recommended over mental calculations. This guide implements the professional methodology outlined by the S Chand Academy to translate complex narrative descriptions into clear family trees.

### The Standard Symbol Notation System

To eliminate ambiguity during calculations, we represent gender and relationship dynamics using a standard set of symbolic conventions:

- **Male Representation (+):** Every male individual is denoted with a plus (+) sign adjacent to their name or variable (e.g., A+).
- **Female Representation (-):** Every female individual is denoted with a minus (-) sign adjacent to their name or variable (e.g., B-).
- **Generational Levels (Vertical Axis):** Parents and older generations are placed vertically above their children. For instance, if A is the father of B, A+ is placed on a higher row with a vertical connecting line leading down to B.
- **Siblings & Same-Level Relations (Horizontal Axis):** Brothers and sisters share the same horizontal row. If B is the sister of J, they are drawn side-by-side (e.g., B- and J).
- **Spousal Relations (Double Bond/Equal Sign):** Husband and wife are placed on the same level, linked by a horizontal double line or specific bond to indicate marriage (e.g., A+ = F-).

### Standard Relationship Glossary

Below is a quick reference glossary of standard English relational terms and their precise definitions as used in competitive logical reasoning:

| English Relation Statement | Equivalent Single-Word Relation |
|---|---|
| Mother's or Father's Son | **Brother** |
| Mother's or Father's Daughter | **Sister** |
| Mother's or Father's Brother | **Uncle (Maternal / Paternal)** |
| Mother's or Father's Sister | **Aunt (Maternal / Paternal)** |
| Mother's or Father's Mother | **Grandmother** |
| Mother's or Father's Father | **Grandfather** |
| Son's Wife | **Daughter-in-Law** |
| Daughter's Husband | **Son-in-Law** |
| Husband's or Wife's Sister | **Sister-in-Law** |
| Husband's or Wife's Brother | **Brother-in-Law** |
| Brother's or Sister's Son | **Nephew** |
| Brother's or Sister's Daughter | **Niece** |
| Uncle's or Aunt's Son/Daughter | **Cousin** |

# Section 2: Exercise 5A - Pointing & Portrait Type Questions

Pointing and Portrait questions involve one person describing their relationship to another individual indirectly. The secret to cracking these questions is to begin decoding backwards from the possessive pronoun 'My' or 'My own' to establish a fixed local reference, then link it to the subject.

## Question 1 (Ex. 5A, Q10 - IBPS 2002)

**Statement:** *Pointing to Kapil, Shilpa said, "His mother's brother is the father of my son Ashish."*

A. Sister-in-law **B. Nephew** C. Niece D. Aunt

**Step-by-Step Family Tree Analysis:**

- Identify the Speaker: Shilpa is the speaker. Since she is referred to as female, we denote her as Shilpa (-).
- Start with 'My': Shilpa says, 'My son Ashish'. So we draw Shilpa (-) and a downward level to Ashish (+).
- Identify 'Father of my son': The father of Shilpa's son must be Shilpa's husband. We draw her husband as a male (+) spouse next to Shilpa (-).
- Link to Kapil: Shilpa says, 'His [Kapil's] mother's brother is [Shilpa's husband]'.
- This means Kapil's maternal uncle (mother's brother) is Shilpa's husband.
- Consequently, Kapil's mother is the sister of Shilpa's husband.
- Since Kapil is the son of Shilpa's husband's sister, Kapil is the Nephew of Shilpa's husband, and thereby the Nephew of Shilpa.

**Correct Answer: B. Nephew**

## Question 2 (Ex. 5A, Q28)

**Statement:** *Pointing to a photograph of a girl, a person said to his friend, "She is the granddaughter of the elder brother of my father."*

A. Daughter **B. Niece** C. Sister D. Aunt

**Step-by-Step Family Tree Analysis:**

- Establish the Base: The person speaking is 'P'. P's father is 'Father' (one level up).
- Identify the Relative: 'The elder brother of my father' is P's Uncle (Tauji). Uncle is on the same generation level as P's father.
- Identify the 'Granddaughter': The granddaughter of P's Uncle would be the daughter of P's cousin.
- Generational Shift: Since the girl in the photograph is the granddaughter of P's father's brother, she is situated one generation level below P.
- Relationship Conclusion: A female relative one generation level below is a Niece.

**Correct Answer: B. Niece**

## Question 3 (Ex. 5A, Q35)

**Statement:** *Looking at a photograph, a lady said to Pramod, "I am the only daughter of this lady and her son is your maternal uncle."*

A. Mother B. Sister **C. Wife** D. Daughter-in-law

**Step-by-Step Family Tree Analysis:**

- Establish Speaker's Origin: The speaker is a Lady (L-). She says, 'I am the only daughter of this lady [in the photograph]'. This means the lady in the photo is the mother of L.
- Analyze the Maternal Uncle: The photo lady's son is Pramod's maternal uncle (Mama).
- Connect the Siblings: Since the lady in the photograph has only one daughter (which is L herself), any son of the photo lady must be L's brother.
- Identify the Mother: The brother of L is Pramod's maternal uncle. The only person whose brother is Pramod's maternal uncle is Pramod's mother.
- Therefore, the speaker (L) is Pramod's mother.
- Identify the Target: The question asks how the speaker is related to Pramod's father.
- Conclusion: Since she is Pramod's mother, she must be the Wife of Pramod's father.

**Correct Answer: C. Wife**

## Question 4 (Ex. 5A, Q50)

**Statement:** *Pointing to a man, a woman said, "This man's son's sister is my mother-in-law."*

A. Daughter B. Wife **C. Grandson** D. Sister-in-law

**Step-by-Step Family Tree Analysis:**

- Deconstruct the Man's Side: 'This man's son's sister' represents the daughter of the man.
- Identify the Mother-in-Law: This daughter is the woman's mother-in-law (i.e., the mother of the woman's husband).
- Trace to the Husband: Since the mother of the woman's husband is the man's daughter, the man is the maternal grandfather (Nanaji) of the husband.
- Find the Relation: The question asks how the woman's husband is related to the man.
- Conclusion: The husband is the Grandson of the man.

**Correct Answer: C. Grandson**

# Section 3: Exercise 5B - Direct Family Relation Questions

Direct family relation questions present a set of distinct relational facts. These are best solved by systematically constructing a single comprehensive Family Tree using the symbolic conventions (+ for male, - for female).

## Question 5 (Ex. 5B, Q3)

**Relations Given:** *A is the mother of B; C is the son of A; D is the brother of E; E is the daughter of B.*

**Tree Construction Steps:**

- A is the mother of B: We draw A (-) on the upper level, with a vertical line down to B (gender of B is currently unknown).
- C is the son of A: We draw C (+) on the same level as B, connecting to A (-). This means C is B's brother.
- E is the daughter of B: We draw B with a vertical line down to E (-).
- D is the brother of E: We draw D (+) on the same horizontal level as E (-). Since D and E are siblings and E is B's daughter, D is B's son.
- Determine relationship for D: We trace from D up to B, and from B up to A. A is the mother of B, so A is the grandmother of D.

**Question:** *Who is the grandmother of D?*
**Correct Answer: A**

## Questions 6 & 7 (Ex. 5B, Q11 & Q12)

**Relations Given:** *A is the son of B; B's sister C has a son D and a daughter E; F is the maternal uncle of D.*

**Tree Construction Steps:**

- A is the son of B: Draw B on the top level, with a vertical line leading down to A (+).
- B's sister C: Draw C (-) on the same horizontal level as B.
- C has son D and daughter E: Draw D (+) and E (-) on the lower level, vertically descending from C (-). This makes D and E siblings.
- F is the maternal uncle of D: Maternal uncle means mother's brother. Since C is the mother of D, her brother is F (+). Draw F (+) on the same horizontal level as B and C.
- This establishes B, C, and F as siblings (B's gender remains unspecified, while C is female and F is male).

**Question 11 (Q11):** *How is A related to D?*
**Correct Answer: Cousin** (Since they are children of siblings B and C).
**Question 12 (Q12):** *How is E related to F?*
**Correct Answer: Niece** (Since E is the daughter of F's sister C).

## Questions 8 & 9 (Ex. 5B, Q21 & Q22)

**Relations Given:** *A is the father of C, but C is not his son; E is the daughter of C; F is the spouse of A; B is the brother of C; D is the son of B; G is the spouse of B; H is the father of G.*

**Tree Construction Steps:**

- A is the father of C, but C is not his son: This implies C is a female. Draw A (+) on the upper level, and C (-) below A.
- E is the daughter of C: Draw E (-) below C.
- F is the spouse of A: Since A is male (+), F must be female (-). They are spouses: A+ = F-. This also means F is the mother of C.
- B is the brother of C: Draw B (+) on the same level as C (-). A and F are B's parents.
- G is the spouse of B: Since B is male, G is female (-). B+ = G-.
- D is the son of B: Draw D (+) below B and G.
- H is the father of G: Draw H (+) on the generation level above G.

**Question 21 (Q21):** *Who is the grandmother of D?*
**Correct Answer: F** (D's father is B, and B's mother is F. So F is the paternal grandmother of D).
**Question 22 (Q22):** *Who is the son of F?*
**Correct Answer: B** (The children of the couple A and F are C (daughter) and B (son)).

## Question 10 (Ex. 5B, Q5-8 Complex Puzzle)

**Relations Given:** *In a family of six persons (A, B, C, D, E, F): C is the sister of F; B is the brother of E's husband; D is the father of A and grandfather of F. There are two fathers, three brothers, and a mother in the family.*

**Tree Construction & Numerical Constraint Checklist:**

- Generational Structure: D is the grandfather of F, so there are three generations: Gen 1 (D), Gen 2 (parents), Gen 3 (F and sister C).
- D is the father of A: This places D (+) on Gen 1, and A (+) on Gen 2.
- C is the sister of F: Draw C (-) and F on Gen 3. Since C is the sister of F, they are siblings.
- B is the brother of E's husband: Since A is on Gen 2 and is male, A must be E's husband. Draw E (-) as A's wife on Gen 2. B (+) is the brother of A, so B is also on Gen 2.
- Check Mother Constraint (1 Mother): E is the mother of Gen 3 (C and F). (Matches!)
- Check Father Constraint (2 Fathers): D is the father of A and B; A is the father of C and F. Total = 2 fathers. (Matches!)
- Check Brother Constraint (3 Brothers): B is a brother (to A); A is a brother (to B); F must be a brother (to C) to satisfy the 'three brothers' count. Thus, F is male (+). (Matches!)

**Question:** *Who is E's husband?*
**Correct Answer: A** (A is married to E, and his brother is B).

# Section 4: Exercise 5C - Coded Blood Relation Questions

Coded Blood Relations substitute relative terms with mathematical operators or special symbols. The most efficient way to tackle these questions in exams is to use Option Elimination based on Gender Requirements and Generational Gap levels, before drawing full family trees.

## Question 11 (Ex. 5C, Q1)

**Directions:** Symbols: P x Q means P is the son of Q; P - Q means P is the wife of Q.

**Given Equation / Target:** **P x Q - S**

A. S is the wife of P **B. S is the father of P** C. P is the daughter of S D. Q is the father of P

**Step-by-Step Symbolic Decoding:**

- Deconstruct P x Q: This translates to: 'P is the son of Q' (P is a male (+) at the generation level below Q).
- Deconstruct Q - S: This translates to: 'Q is the wife of S' (Q is female (-), S is male (+), and they are spouses).
- Synthesize: Since P is Q's son, and Q is married to S (husband), P must be S's son as well.
- Verify Options: Therefore, S is the father of P.

**Correct Answer: B. S is the father of P**

## Question 12 (Ex. 5C, Q2)

**Directions:** Symbols: Same as above (P x Q means P is son of Q; P - Q means P is wife of Q).

**Given Equation / Target:** **T - S x B**

A. S is the husband of T B. T is the daughter-in-law of B C. S is the son of B **D. B is the mother of S**

**Step-by-Step Symbolic Decoding:**

- Deconstruct T - S: 'T is the wife of S' (T is female (-), S is male (+)).
- Deconstruct S x B: 'S is the son of B'. B's gender is currently unknown.
- Synthesize: Since S is the son of B, and T is S's wife, T is indeed the daughter-in-law of B. S is the husband of T, and S is B's son.
- Evaluate for Statement That is NOT True: Option D claims 'B is the mother of S'. Because B's gender is unspecified (B could be father or mother), we cannot definitively state B is the mother. Thus, Option D is not necessarily true.

**Correct Answer: D. B is the mother of S (Not definitively true)**

## Question 13 (Ex. 5C, Q3)

**Directions:** Symbols: A / B means A is the daughter of B; A + B means A is the father of B.

**Given Equation / Target:** **P / R + S + Q**

A. P is the daughter of Q **B. P is the aunt of Q** C. P is the sister of Q D. Q is the mother of P

**Step-by-Step Symbolic Decoding:**

- Deconstruct P / R: 'P is the daughter of R' (P is female (-) one level below R).
- Deconstruct R + S: 'R is the father of S' (R is male (+) one level above S). This implies P and S are siblings (P is the sister of S).
- Deconstruct S + Q: 'S is the father of Q' (S is male (+) one level above Q).
- Synthesize: Since P is the sister of S, and S is the father of Q, P is the father's sister of Q.
- Conclusion: Father's sister is defined as Aunt.

**Correct Answer: B. P is the aunt of Q**

## Question 14 (Ex. 5C, Q4)

**Directions:** Symbols: A - B means A is the wife of B; A + B means A is the father of B.

**Given Equation / Target:** **P - R + Q**

A. P is the daughter of Q  B. P is the sister of Q  **C. P is the mother of Q**  D. Q is the mother of P

**Step-by-Step Symbolic Decoding:**

- Deconstruct P - R: 'P is the wife of R' (P is female (-) married to R (+)).
- Deconstruct R + Q: 'R is the father of Q'.
- Synthesize: Since R is the father of Q, and P is married to R, P is the mother of Q.

**Correct Answer: C. P is the mother of Q**

## Question 15 (Ex. 5C, Q5)

**Directions:** Symbols: P @ Q means P is the husband of Q; Q $ R means Q is the mother of R; R # M means R is the father of M.

**Given Equation / Target:** **P @ Q $ R # M**

A. P is the maternal grandfather of M  **B. P is the paternal grandfather of M**  C. M is the grandson of P  D. P is the father of M

**Step-by-Step Symbolic Decoding:**

- Deconstruct P @ Q: 'P is the husband of Q' (P+ = Q-).
- Deconstruct Q $ R: 'Q is the mother of R' (R is on the level below Q. Since P is married to Q, P is the father of R).
- Deconstruct R # M: 'R is the father of M' (M is on the level below R, making R a male (+)).
- Synthesize: P is the father of R, and R is the father of M. This makes P the grandfather of M.
- Determine Subtype: Since R is a male (+), P is related to M through M's father (R). Therefore, P is the paternal grandfather of M.

**Correct Answer: B. P is the paternal grandfather of M**

## Question 16 (Ex. 5C, Q6 - Complex Equation Finding)

**Directions:** Symbols: A % B means A is the daughter of B; A @ B means A is the husband of B; A $ B means A is the mother of B.

**Given Equation / Target:** **Target: Show that R is the sister of H.**

A. R $ D @ F % H **B. R % D @ F $ H** C. R @ D % F $ H D. H % D @ F $ R

**Step-by-Step Symbolic Decoding:**

- Determine Gender Constraint: For R to be a sister of H, R must be female (-).
- Eliminate Options: In Option C, we see 'R @ D' which would mean R is the husband of D (male), which is impossible. We eliminate C.
- Examine Option B (R % D @ F $ H):
- - 'R % D' means R is the daughter of D (R is female (-) and D's child).
- - 'D @ F' means D is the husband of F (D is male (+), F is female (-), and F is the mother of R).
- - 'F $ H' means F is the mother of H.
- Synthesize: Since F is the mother of both R and H, and R is a female, R is the sister of H. This perfectly satisfies the target relationship.

**Correct Answer: B. R % D @ F $ H**